

Deliverable 9 – Power BI Dashboard Development Report

Healthcare Patient Experience Analytics using Power BI

Introduction

The Healthcare Patient Experience Analytics solution was developed using **Microsoft Power BI** to transform healthcare survey data into an interactive Business Intelligence platform. The dashboard enables healthcare executives, policy makers, quality officers, and state administrators to monitor patient satisfaction, evaluate healthcare performance, benchmark results, and identify improvement opportunities through dynamic visualizations.

The solution demonstrates enterprise-level reporting by combining interactive dashboards, DAX measures, an optimized data model, and user-friendly visualizations.

Project Overview

The dashboard solution consists of **six interactive reports**, each designed to address a specific business objective.

Report	Business Purpose
Executive Overview	Provides a high-level summary of national healthcare performance.
State Performance	Evaluates patient satisfaction across individual states.
Patient Experience Measures	Analyzes healthcare quality measures and patient experience indicators.
National vs State Benchmark	Compares state performance with national averages.
Regional Performance	Evaluates healthcare performance across U.S. regions.
Trend Analysis	Monitors patient satisfaction trends across reporting periods.

Dashboard Features

Executive Overview Dashboard

Objective

Provide senior management with a comprehensive overview of national healthcare performance.

Key Features

- KPI Cards
 - National Satisfaction Trend
 - Filled Map
 - Treemap
 - Matrix
 - Interactive Filters
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State Performance Dashboard

Objective

Analyze healthcare performance at the state level.

Key Features

- State Ranking
 - State Satisfaction Score
 - Regional Comparison
 - Measure Performance Analysis
 - Scatter Chart
 - Matrix
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Patient Experience Measures Dashboard

Objective

Evaluate patient satisfaction across healthcare quality measures.

Key Features

- Measure Ranking
- Top Performing Measures
- Lowest Performing Measures
- Ribbon Chart

- Treemap
 - Trend Analysis
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National vs State Benchmark Dashboard

Objective

Compare state healthcare performance against national benchmarks.

Key Features

- National Average
 - State Average
 - Variance Analysis
 - Benchmark Comparison
 - Waterfall Chart
 - Combo Chart
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Regional Performance Dashboard

Objective

Compare healthcare performance across regions.

Key Features

- Regional Ranking
 - Regional Satisfaction Score
 - Top 10 States
 - Bottom 10 States
 - Regional Trend Analysis
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Trend Analysis Dashboard

Objective

Monitor changes in patient satisfaction over time.

Key Features

- Satisfaction Trends
- Reporting Period Comparison
- Measure Trends
- Response Rate Trends
- Improvement Analysis

Visualizations Used

The dashboard includes a variety of visualizations selected to communicate healthcare insights effectively.

Visualization	Business Purpose
KPI Cards	Display important performance indicators
Filled Map	Visualize state-wise patient satisfaction
Matrix	Detailed comparison of healthcare measures
Clustered Bar Chart	Compare performance across categories
Clustered Column Chart	Analyze reporting periods
Line Chart	Display trends over time
Treemap	Compare healthcare quality measures
Ribbon Chart	Show ranking changes
Scatter Chart	Compare state performance
Waterfall Chart	Analyze variance and improvement

These visualizations align with the project requirements for enterprise-quality reporting. :contentReference[oaicite:1]{index=1}

Interactive Features

The dashboard incorporates interactive features to improve user experience and support exploratory analysis.

Feature	Purpose
Slicers	Filter reports by state, region, reporting period, and quality measure
Drill-down	Analyze data at multiple levels of detail
Drill-through	Navigate to detailed report pages
Report Tooltips	Display additional information on hover
Sync Slicers	Maintain consistent filtering across reports
Conditional Formatting	Highlight significant performance variations

Dashboard Design Principles

The dashboards were designed following Business Intelligence best practices.

- Consistent healthcare-themed color palette.
 - Clean and professional layout.
 - Clear visual hierarchy.
 - Interactive filtering.
 - Appropriate chart selection.
 - Minimal visual clutter.
 - Consistent typography.
 - Responsive report design.
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Business Value

The dashboard provides the following business benefits:

- Centralized healthcare performance monitoring.
 - Real-time KPI tracking.
 - National and state benchmarking.
 - Regional performance comparison.
 - Trend analysis across reporting periods.
 - Identification of improvement opportunities.
 - Support for evidence-based healthcare decisions.
 - Executive-ready reporting through interactive dashboards.
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Dashboard Summary

Component	Count
Dashboard Pages	6
KPI Measures	14+
Interactive Filters	Yes
DAX Measures	Yes
Star Schema	Yes
Power Query Transformations	Yes
Interactive Visualizations	25+

Conclusion

The Healthcare Patient Experience Analytics dashboard successfully delivers an enterprise-level Business Intelligence solution using Microsoft Power BI. Through interactive visualizations, optimized DAX measures, and a well-structured data model, the dashboard enables stakeholders to analyze patient satisfaction, benchmark healthcare performance, monitor trends, and make informed decisions that contribute to continuous improvements in healthcare quality.

